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Professor Matthew J. Rosseinsky OBE FRS & Recruitment Team

Department of Chemistry, Materials Innovation Factory

University of Liverpool

Liverpool, UK

Dear Professor Rosseinsky and the Recruitment Team,

I am writing to express my keen interest in the **Postdoctoral Research Associate in Computationally-Driven Materials Design and Discovery** position (Job Ref: 114193) at the **University of Liverpool**. I am currently completing my PhD under the supervision of Prof. Dario Alfè at University College London (thesis submitted, degree expected October 2026). I have developed broad expertise in first-principles calculations, atomistic modelling, machine learning, agentic AI systems for workflow automation, and AI-driven materials discovery, and I believe these skills align well with your project.

My core PhD research involves developing ab-initio methods for the phase diagrams of binary alloys, combining DFT and molecular dynamics simulations to train and validate machine-learning inter-atomic potentials, alongside thermodynamic integration and free-energy calculations within a fully automated workflow. Alongside my thesis, I have contributed to collaborative projects with experimentalists on material defects, electrides, and electrocatalytic materials for hydrogen production, which involve strain-tuned band-edge alignment and Gibbs free-energy pathway calculations to identify optimal catalytic conditions. Prior to my PhD, during my Master's, I worked on the electronic, structural, mechanical, magnetic, optical, and quantum-transport properties of materials using DFT and NEGF approaches. I was then selected as a Research Associate at Khalifa University, UAE, where I worked on high-throughput computational discovery and investigated photocatalytic applications.

In addition to my research, I have been a Teaching Assistant across three UCL courses. In this role, I supervised a master's student project, ran practicals in DFT, machine learning, and deep learning, and deployed deep learning models for student outreach.

My research has led to more than a dozen peer-reviewed publications, most as first and corresponding author, presented at national and international conferences, and I serve as a peer reviewer for several journals in the field.

The Rosseinsky Group's pioneering research within the ERC Advanced Grant MaterED project and the Materials Innovation Factory broadly intersects with my background. In particular, your focus on structure-driven discovery combining first-principles calculations, machine-learned potentials, and automated reasoning to predict functional inorganic materials strongly resonates with my research expertise and interests. I am confident that this position offers an exceptional opportunity to contribute to your digital materials design workflows and collaborate closely with your experimental teams. I look forward to discussing my application further.

Yours sincerely,

Shambhu Bhandari Sharma